Analog Mixing Console with Digital Control and Automated Signal Path Configuration

Capstone Project Proposal

Skylar Denno

Evan Forcucci

Daniil Petrov

Xinhao Chen

Daniel Salas

Luke Garrity

Advisor: Milad Siami [**Team #2**]

Northeastern University

Department of Electrical and Computer Engineering

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1 Abstract

An analog mixing console cannot remember its own settings. Every parameter lives in the position of a potentiometer, and the audio passes through that potentiometer, so the two roles cannot be separated. Restoring a session means turning every knob by hand against a paper recall sheet, and the result is never exact. A digital console solves recall completely, but only by converting the audio to numbers at the input and back to a voltage at the output.

This proposal describes a console that takes neither path. The audio stays analog from the input connector to the summing bus. Only the control values become digital. A rotary encoder produces a number instead of a resistance, a control processor stores that number, and a control voltage carries it to the analog circuit. The circuit then responds as though a hand had moved a knob. Voltage controlled amplifiers set every gain, and operational transconductance amplifiers supply every variable resistance in the equalizer. One command therefore reconfigures a whole channel, and nothing in the signal path moves.

The design covers 16 channels and a master section. One STM32G474 processor serves each group of four channels through four independent I^2C buses, and it drives 20 control voltages and 42 input pins per channel. Each channel carries a microphone preamplifier or a line receiver, a four band state variable filter equalizer, a compressor with a digital detector and an analog gain element, auxiliary sends, a pan control, and a motorized fader with a touch sensor.

The difficulty is not the control plane. The difficulty is keeping the audio clean while converters, expanders, shift registers, motor drivers, and switching supplies all sit centimetres from the signal path. Three quantities decide the outcome: the linearity of every voltage controlled element, the noise that the control plane couples into the audio, and the repeatability of a recall as the temperature changes. Success is therefore a measurement rather than an opinion. The project reports total harmonic distortion, the noise floor, the crosstalk between channels, and the recall error across a temperature range, and it compares each figure against a conventional analog desk.

2 Introduction

A mixing console routes, shapes, and balances many audio signals at once, before summing each processed signal together into a full mixed, recorded performance of a song. Two kinds of audio mixing console serve this job today, and each one gives up something that the other holds.

An analog console carries the audio as a continuous voltage from the input to the output. Many audio engineers prefer the result of analog hardware and insist, even today, to use it despite its expense and many drawbacks. These drawbacks include the issue of recall of settings. An analog console cannot store a setting to load later, since all the knobs and faders control physical potentiometers that have physical resistances that the audio signal passes through. To set a parameter (a resistance) for a different configuration or song, one must physically rotate the knobs by hand to physically recall the state. To reconfigure an entire console (often with 64+ channels, each with several tens of knobs), can take hours. The state of a session lives in the position of every knob and every fader on the surface. An audio engineer who needs that state again writes it (or often their assistant writes it) on a paper recall sheet and rebuilds it by hand. The rebuild is slow, and it is never exact.

Conversely, a digital console converts the audio to discrete samples at the input and back to a continuous voltage at the output. All the processing is done digitally with digital signal processing (DSP). The digital console stores and recalls every parameter without error, but the audio only reaches those settings and that memory after passing through converters and arithmetic. No matter if DSP really "sounds" worse than analog processing or not, the analog music equipment market is massive.

We propose a mixing console that takes a third path. The audio stays analog through the whole console signal chain. Only the control values become digital, and using deep integration between digital and analog systems, we may configure the console's analog signal path instantly and exactly, eliminating the need for manual recall.

What the Console Does

In traditional mixing consoles, the potentiometers do two jobs at once, and these jobs are inseparable. They provide a physical control and indication of position, and they provide a variable resistance related to that position. If we separate the potentiometer into a rotary encoder, which provides the physical control and indication, but no resistance, and a voltage controlled resistor of some sort, which provides the variable resistance without moving parts, we can eliminate the problem of having to physically match the state on real resistor strips for recall.

In our console, every parameter that an engineer would normally set with a potentiometer becomes a number in the memory of a control processor. That number is set by a rotary encoder, and that number reaches the analog circuit as a control voltage, and the circuit responds as if a hand had moved a knob.

Crucially, the controller itself can also change all of the control voltages independently of the physical positions of any knobs, instantly reconfiguring hundreds of parameters from a saved state with no moving parts. Additionally, an arbitrary number of controls may be automated to change values throughout a song, and arbitrarily many parameters may be changed at once in real time for live applications (unlike a physical potentiometer, where an audio engineer only has two hands).

An audio console is simply made up of many individual channels of identical signal processors. Each channel processes each part/layer of audio within of the song, such as the snare drum, the bass guitar, the lead vocal, etc. and each layer of audio needs different processing to fit in the full "picture" of the full recorded performance. Audio consoles simply combine many of these channels together, add some useful routing options, and sum the processed outputs of each channel together at the end. We aim to create a console with a modest 16 channels, using one STM32 microcontroller for every group of 4 channels. The actual remote control of parameters and analog processing of audio happens within those channels, and each channel identically requires the following features.

Every channel carries its own microphone preamplifier or a line receiver, its own four band equalizer, a dynamic range compressor, routing options for auxiliary sends, a pan control, and a fader / main volume control. For our design, we plan for voltage controlled amplifiers (VCAs) to set every gain in that chain. Operational transconductance amplifiers (OTAs) will supply every variable resistance, which are needed to control the corner frequencies and the Q (quality/BW) factor of each band in the equalizer. Analog multiplexers select the band shapes, the bypass paths, and other routing options. Section 4 describes each of these circuits in turn.

The control surface reports the state that the memory holds. Each channel carries ten rotary encoders with rings of LEDs around each encoder to show their "position" in a way that can be changed remotely without movement. Each channel also will have sixteen buttons with lamps, a small display, and a motorized fader with a touch sensor. None of these controls touch the audio. An encoder produces a number and nothing else, so the same encoder can serve the equalizer, the compressor, or the sends as the engineer selects. One physical control therefore covers several parameters, and the panel holds far fewer controls than the channel has settings, which reduces the cost, complexity, and size as compared to a fully analog console. Additionally all the rotary encoders may be the same across the entire system, whereas using potentiometers in a fully analog audio path usually requires a large variety of hard-to-source potentiometers with different values and exotic taper curves.

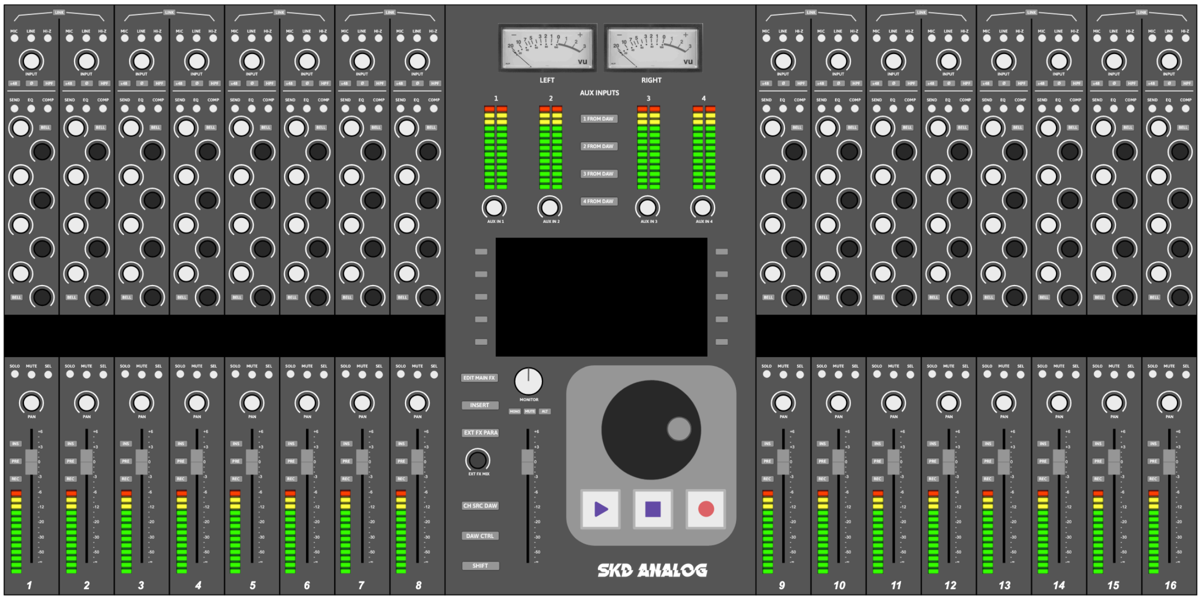

Figure 1: Mockup of the front panel of a 16 channel console under this vision.

Greatest Challenges

Moving the control plane into a processor is not the hard part. Keeping the audio clean while that processor sits centimeters away, along with the depth and number of points of the analog/digital integration, is the hard part. Three problems decide whether the approach works.

The first problem is linearity. A fixed resistor and a good operational amplifier, as used in traditional consoles, are close to ideal. A voltage controlled element is not. Every gain stage and every filter in the channel therefore could add distortion that a passive component would not add. The differential pair inside an operational transconductance amplifier stays linear only across a few tens of millivolts, so the design attenuates the signal before it reaches that input and takes advantage of linearizing distortion, which predistorts the signal in a way such that the OTAs nonlinearities will undo this distortion.

The second problem is noise, of which the digital control circuitry supplies the largest source. A digital console can place its processor far from any analog circuit, and the digital audio is impervious to external interference. This console does not have this luxury. Converters, port expanders, shift registers, motor drivers, and switching supplies all sit near the signal path. Noise on a control port also does more damage than noise on a signal wire, because a voltage controlled amplifier multiplies that noise into the audio. One millivolt on a control port for the filter circuitry produces sidebands only 55 dB below the signal.

The third problem is recall accuracy. For the most part, the console runs open loop. No sensor measures the true gain of a channel, and no controller corrects it. This is in order to allow the audio engineer to change anything in realtime, whenever they want. We may add a short startup sequence that calibrates some things, but for normal operation, there is nothing on the output side to ensure nothing drifts. The transconductance of an operational transconductance amplifier and the gain constant of a voltage controlled amplifier both scale with absolute temperature. A stored number therefore does not produce the same result on a cold morning that it produced during the session it came from. Section 4.3 discusses this topic in more detail.

Goal

The goal of this project is first and foremost a demonstration. We aim to show that our audio console can hold every parameter in memory, recall it and reconfigure a fully analog signal path on command, and still meet the audio performance that engineers expect from a conventional analog desk. Digital control must cost nothing that even an experienced audio engineer can hear.

Additionally, a second and more movable goal is to build these concepts into a somewhat polished, nearly release-ready "demonstration product." Since much of the challenges with noise and linearity can only solved by designing our own PCBs, and since some of the digital systems will be too fast to rely on breadboards, due to their intrinsic parasitic capacitance and inductance, we will have to design PCBs and use hobbyist/prototype manufacturing services such as JLCPCB. If we just make a nice enclosure, making a "proto-product" kind of thing is not much of a step past demonstration.

However, for the first (and main goal of demonstration), success is a measurement and not an opinion. We will report total harmonic distortion, the noise floor, the crosstalk between channels, and the repeatability of a recall across a range of temperature. We will compare these values to existing, fully analog tradition, processors. A design that recalls the settings perfectly but adds audible distortion or hiss has failed the claim that matters.

2.1 Applications in Other Fields

The split between an analog signal plane and a digital control plane is neither new nor unique to audio. Numerous industries run a physical processes that produce and handle highly sensitive analog signals in their native domain and command those processes from a computer. They meet the same problems that this console meets, so their answers inform its design.

Electric power gives the clearest case, because nobody can digitize the signal. A substation handles hundreds of megawatts, and if a converter could sample that power and rebuild it, what would power the output buffer?! So the power path stays analog by necessity. Every improvement must therefore come from the control plane, and the IEC 61850 standard [1] carries settings, measurements, and commands across that plane.

The devices in a substation divide the same way the channel strip on an audio console divides. The on-load tap changer of a transformer moves in discrete steps, much like the switched resistor network inside the PGA2500 [2] digitally controlled microphone preamplifier we will use. A static VAR compensator varies its output in a continuous way through a thyristor firing angle, much like a voltage controlled amplifier follows a control voltage. Recall appears in the same form as well. A protective relay holds several groups of settings, and an operator selects a different group when the network configuration changes. One command therefore moves many parameters to a known state, which is exactly what a snapshot does on a console.

A power system also adds one rule that this design needs. An interlock checks the preconditions before a command executes, because the wrong order of operations can damage infrastructure. The recall sequence of Section 4.3 follows the same logic when it mutes a channel before it writes new values, to avoid clicks and pops on the output, which when amplified could (and have, many times) destroy the speaker drivers in highly accurate studio monitors with extremely fast transient responses.

Two further fields each contribute a method. A chemical plant stores its settings as recipes and recalls them at the start of a production run, which is a snapshot under a different name. Process control also names the transition itself. A bumpless transfer moves an output to a new setpoint without a step at the process, and the mute and ramp of a console recall performs the same work. A phased array radar holds thousands of antenna elements, and no two of them match. Every element therefore carries a calibration table against frequency and against temperature. Section 4.3 applies that same method to each channel of this console, and for the same reason.

3 System Overview

3.1 Analog Systems

Remote Controlled Analog Systems

To remotely control parameters of the analog signal processing, all the required processing may be (and usually is, even in fully analog systems) designed to only require variable resistors and variable gain. For example, to change the resonant frequency of a band on the equalizer, we may keep the capacitor value constant and vary only the resistor to change the resonant frequency. The implementation details of these voltage controlled amplifiers and voltage controlled resistors are covered in Sections 4.1 and 4.2, respectively.

In short, voltage controlled amplification will be implemented using voltage controlled amplifiers (VCAs), specifically the SSI2164 [3] for most signals, and the PGA2500 digitally controlled microphone preamplifier (on an SPI control bus rather than using an analog control voltage) for the microphone inputs. Any switching or reconfiguration of channel paths will be implemented by the DG409 [4] CMOS analog multiplexer. Voltage controlled resistors will be implemented using the LM13700 [5] operational transconductance amplifier (OTA).

For these components, temperature often changes their behavior, and parameters may differ between units. Details on calibration and temperature compensation schemes are outlined in Section 4.3. Using these components, a voltage controlled parametric equalizer (Section 4.4) may be implemented.

Control Voltages

To produce the control voltages, we will rely on a set of MCP4728 [6] digital-to-analog converter (DAC) chips on I2C busses (running at "Fast mode," 400kHz [7]). These chips have 12-bit accuracy (4,096 voltage levels), which provides more than enough definition for our purposes. We will use a low sample rate around 400Hz, as these control voltages do not need fast changes or high bandwidth. The 0-3.3V output from these DACs will be mapped to a larger voltage range of around $+/-15V$ using low offset op-amps and a reference offset voltage. The amplification circuit will also provide low pass filtering which both smooths the control voltages and removes any digital noise. Only then will the control voltages be allowed to pass physically to the low noise analog section. More information on this topic is in Section 4.9.

Noise and Power

The analog systems will be powered by a set of low noise, bipolar rails at about $+/-16.5V$. The power supply for the analog components will be of a linear topology, to avoid polluting the voltage rails with the switching noise present in other power supply designs. Additionally, the DACs and ADCs will use a dedicated 3.3V voltage reference, designed as a small, low noise supply and separate from the main, high current 3.3V logic supply, but sharing the same ground potential. The power supply is discussed in Section 4.10.

To further isolate the analog section from the digital section, the analog circuitry will be physically located on a different printed circuit board (PCB) than the digital components. The boards will be the same size and stack on top of each other in a mezzanine configuration, with pin headers, similarly to a shield board for microcontroller development boards, but at a much larger size. The digital PCB will be on top of the analog PCB. Any high speed or low-noise analog signals that must travel between the boards will do so via the pin headers. Interlacing these signals with ground pins in the physical pin header layout will reduce crosstalk between channels. We will avoid placing high speed or low noise signals near each other in the pin header, without a ground pin between them.

The digital and analog systems will essentially share the same ground node, but the digital and analog grounds will only connect at a single point at the power supply. Most of the noise protection will be due to the PCB layout and physical separation, rather than trying to separate the ground nodes. The digital PCB, from top to bottom, will likely have the four following copper layers: Top: Digital signals, SMD; Inner 1: Ground plane; Inner 2: Digital signals (heavily shielded); Bottom: Ground plane. Stitching vias in the ground plane will help equalize them with a very low impedance path between them. Ground vias will also be placed near each chip's ground pins. The analog PCB (below the digital PCB), from top to bottom, will have the following four copper layers: Top: Analog signals, SMD; Inner 1: Ground plane; Inner 2: Ground plane; Bottom: Analog signals. With this stackup, we can see that the bottom layer of the digital PCB, sitting right above the first layer of the analog PCB, is a ground plane, rather than a signal layer. This will help prevent interference from the digital electronics from inducing noise in the sensitive analog circuitry.

Local decoupling and filtering will be added physically closer to many of the chips and devices to shunt any noise from crosstalk of digital traces and systems to ground.

Summing Bus

The output of each of the individual audio channels will feed a summing bus, from which the "mixed" audio may be recorded from back into a computer. The processing and gain/attenuation happens on all the individual channels, and the summing bus is just a "dumb" summing amplifier with equal input impedance see by every channel output (post channel gain).

3.2 Digital Systems

STM32G474

For this project, the STM32 microcontroller was chosen. Specifically, we chose the ARM based STM32G474 [8] variety, for its expansive I/O, deep analog integration with onboard DACs and ADCs, and its ability for standard-speed USB. We plan to run the CPU clock at its maximum speed of 170MHz. This will be more than enough processing power for our requirements. Additionally, our team already has a few Nucleo 64 development boards of this exact chip, allowing those to be excluded from the budget.

We will use an external 24MHz crystal oscillator on our final PCBs (exact same as the Nucleo boards) to simplify all of the different timers for busses running at different speeds. If the firmware will not fit on the chip's internal flash, we will add an SPI flash chip, on a separate SPI bus. This connection, while slow in today's standards, will be more than enough for the controller to access the program during operation.

Channel Buckets

Each STM32 controller will be in charge of a "bucket" of four channels. We will make everything for this entire project as a set of 4 channels, and during final production, we will combine a few of these buckets together to create a console. The current goal is 16 channels, requiring 4 sets of 4-channel buckets. The numbers listed in the budget proposal are for this 16 channel configuration. However, if budget requires, the design can be reduced to 12 or 8 channels extremely easily, as a result of this bucket setup.

Master Section

There will also be a simple master section with its own STM32G474. The master section does not need remote control, as all the processing will be done on the individual channels of audio. The master section is just a "dumb" summing bus which accepts the output of an arbitrary number of channels, with their respective gain factors already applied by the VCAs. The summing bus can then be fixed gain for each channel. The master section controller will mostly be to run the large center screen, which will allow for track naming and console setting configuration. This highly modular design allows a console with any number of channels (divisible by 4) to be created by using any number of individual buckets of 4 channels and a single master section.

Communication Between Devices and Pin Assignment

The controllers will communicate with each other on an SPI (Serial Peripheral Interface) bus, but for the most part, communication between controllers is not needed, since each manages the signal path configuration of its own four channels, and the outputs from those channels just dump onto the main "dumb" summing bus with the processing and gain already completed. This bus will be mostly used for saving the console state, where the master section controller polls each of the bucket controllers for their channel parameter values. Each bucket will dump its numerous values of settings onto the bus, and the master controller will collect these and save them. Upon loading, the same thing will happen in reverse.

The STM32G474 has four I^2C (Inter-integrated Circuit) buses and three SPI (Serial Peripheral Interface) buses. Each I^2C bus consists of a clock pin and a data pin, carrying signals in both directions. Multiple devices may communicate, with 3 bit addresses, allowing 8 addressable chips on the same bus. Each I2C bus will be used for one channel's chips, so all four I2C busses will be identical for simplicity, and we can address specific channels by changing which I2C bus we are communicating on.

Each I2C bus will have three, MCP23017 [9] GPIO expanders (giving 42 total input pins per channel = 168 additional GPIOs for each bucket/STM32). These pins will be connected to the physical control devices such as the rotary encoders and pushbutton switches. More information about GPIO expansion and control is in Section 4.8. Each bus will also have five, MCP4728 quad DACs, for a total count of 20 control voltage DAC channels per console channel, which fits in the required number for the analog processors we plan to implement. More information about control voltage DACs is in Section 4.9.

The STM32 controller also has three SPI busses. Bus 1 will be configured as a 4-wire full duplex, in slave mode, where a master controller selects which chip to talk to. This is for the aforementioned communication between controllers and the master section. SPI bus 2 will be a 2-wire transmit only bus, which will be connected to all four of the small channel LCD screens. These screens only receive data and never send it. More information about the LCD screens is in Section 4.7.

Finally, the third SPI bus will be reserved for a potential SPI flash chip, if the onboard flash is not enough to contain the controller's firmware. We may also use dedicated ADC channels for the dynamic compressor sidechain (See Section 4.5) and control voltage envelope generation. We might also move some things around, adding an SPI GPIO expander such as the MCP23S17 to this bus, to move signals like the fader touch sensors or display chip select pins, making room to use onboard ADCs for the compressor instead (See Figure 3).

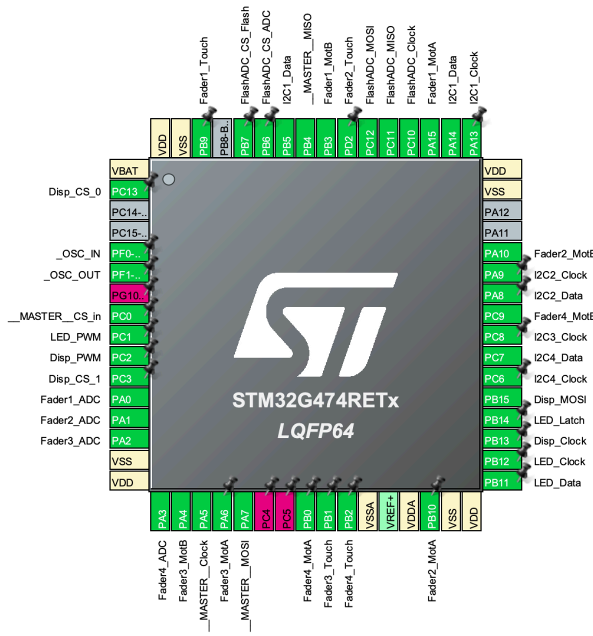

Figure 2: Pin assignments in STM32CubeMX. PG10 is dedicated for reset only. PC4 and 5 are multiplexed with PA2 and 3, respectively on the Nucleo development board, and to reduce complexity, we just avoid PC4 and 5. This is for one bucket of 4 channels. PB8 is the boot mode pin, left open in case we need to tell the microcontroller to boot from the external SPI flash chip (if the firmware does not fit on internal flash)

FN	DESC	PROTO	PC FN	PIN NAME	PIN #	LABEL
Controls 1	MCP23017 GPIO Expansion x4, MCP4728 DAC x4	I2C 1 400kHz	Clock	PA13	49	I2C1_Clock
			Data	PA14	50	I2C1_Data
Controls 2 + CV Xtra	MCP23017 GPIO Expansion x4, MCP4728 DAC x4	I2C 2 400kHz	Clock	PA9	43	I2C2_Clock
			Data	PA8	42	I2C2_Data
CV 1	MCP23017 GPIO Expansion x4, MCP4728 DAC x4	I2C 3 400kHz	Clock	PC8	40	I2C3_Clock
			Data	PB5	58	I2C3_Data
CV 2	MCP23017 GPIO Expansion x8	I2C 4 400kHz	Clock	PC6	38	I2C4_Clock
			Data	PC7	39	I2C4_Data
Comm. To MASTER	Serial Out SPI Serial In SPI	SPI 1 (FullDuplex Slave)	Clock	PA5	19	__MASTER__Clock
			MOSI	PB4	57	__MASTER__MOSI
			MISO	PA7	21	__MASTER__MISO
			Chip Sel In	PC0	8	__MASTER_CS_in
Displays	SPI Clock Serial Out SPI SPI Chip sel bit 0 (multiplex to 4) SPI Chip sel bit 1 (multiplex to 4)	SPI 2 (Trans Only Master)	Clock	PB13	35	Disp_Clock
			MOSI	PB15	37	Disp_MOSI
			Chip Sel [0]	PC13	2	Disp_CS_0
			Chip Swl [1]	PC3	11	Disp_CS_1
Flash / Compressor ADC	SPI Clock Serial Out SPI Serial In SPI SPI Chip sel ADC (4 channel ADC for comp) SPI Chip sel flash (Load prgm)	SPI 3 (FullDuplex Master)	Clock	PC10	52	FlashADC_Clock
			MOSI	PC12	54	FlashADC_MOSI
			MISO	PC11	53	FlashADC_MISO
			Chip Sel ADC	PB6	59	FlashADC_CS_ADC
			Chip Sel Flash	PB7	60	FlashADC_CS_Flash
Fader Position (Onboard ADC)	Read position of slide pots thru built in adcs	ADC 1 Ch 1-4	Analog val	PA0	12	Fader1_ADC
			Analog val	PA1	13	Fader2_ADC
			Analog val	PA2	14	Fader3_ADC
			Analog val	PA3	17	Fader4_ADC
Fader Touch	Sense if faders are being touched (from TTP223-BA6's output)	GPIO Input		PD2	62	Fader1_Touch
				PB9	55	Fader2_Touch
				PB1	25	Fader3_Touch
				PB2	26	Fader4_Touch
FADER MOTOR	Fader motor controller with PWM H bridges (DRV8833 x2)	TIM2 PWM 24khz ch1/2	H-bridge motor ctrl PWM	PA15	51	Fader1_MotA
				PB3	56	Fader1_MotB
		TIM2 PWM 24khz ch3/4	H-bridge motor ctrl PWM	PB10	30	Fader2_MotA
				PA10	44	Fader2_MotB
		TIM3 PWM 24khz ch1/2	H-bridge motor ctrl PWM	PA6	20	Fader3_MotA
				PA4	18	Fader3_MotB
		TIM3 PWM 24khz ch3/4	H-bridge motor ctrl PWM	PB0	24	Fader4_MotA
				PC9	41	Fader4_MotB
LEDs	Shift register out for all LEDs in this bucket of 4 channels	Serial Out Shift	Serial clk shift reg	PB12	34	LED_Clock
			Serial data shift reg	PB11	33	LED_Data
			Latch	PB14	36	LED_Latch
		TIM1 ch2 24khz	PWM	PC1	9	LED_PWM
Display Backlight	Dim backlight for displays	TIM1 ch3 24khz	PWM	PC2	10	Disp_PWM
OSC	High freq external OSC (24MHz crystal)	OSC IO		*PF0*	*5*	*_OSC_IN*
				PF1	*6*	*_OSC_OUT*
USB	MIDI control? Computer control?	USB 2.0 Full Speed	*USB D+ data duplex*	*PA12*	*46*	*USB_DP*
			USB D- data duplex	*PA11*	*45*	*USB_DM*
RESET	Active low (built in pull up)	Rst		*PG10*	*7*	*NRST*
PRGM	Programming firmware to the chip (not on devboard)	SWD	*SWD Data*	*PC4*	*22*	*SWDIO*
			SWD Clock	*PC5*	*23*	*SWCLK*
VSS_Analog	Clean gnd ref for adcs (analog gnd plane)	Pow		*VSSA*	*27*	*VSSA*
VDD_Analog	Clean 3v3 ref for adcs (linear 3v3)	Pow		*VDDA*	*29*	*VDDA*
V_Battery	Tie to 3v3 digital regular VCC (no battery)	Pow		*VBAT*	*1*	*VBAT*
V_reference +	Tie to clean 3v3 same as VDDA	Pow		*VREF+*	*28*	*VREF+*
VDD	Main 3v3 power supply (conn all)	Pow		*VDD*	*16*	*VDD*
				VDD	*32*	*VDD*
				VDD	*48*	*VDD*
				VDD	*64*	*VDD*
VSS	Main GND supply (conn all, decouple etc)	Pow		*VSS*	*15*	*VSS*
				VSS	*31*	*VSS*
				VSS	*47*	*VSS*
				VSS	*63*	*VSS*
Boot Mode	Boot Mode leave free			*PB8*	*61*	*PB8-BOOT0*
NC	No connect			*PC14*	*3*	*OSC32_IN*
				PC15	*4*	*OSC32_OUT*

Figure 3: Pin assignment listing. Some of the general purpose output pins may be moved to extra pins on the MCP23017s instead.

10 pins are configured as pulse width modulation (PWM) outputs. 8 of these are for the fader motor H-bridge control (two for each motor, see Section 4.6). The other two pins are to change the brightness of the LCD backlights and of all the LEDs on the bucket of channels, through PWM on every LED driver shift register's output enable pin (See Section 4.8). To keep potential audible noise out of the audio, all PWM signals will run at 24KHz or more, beyond human hearing.

Additionally (not shown in Figures 2 and 3) we plan, if needed, to have a small temperature sensor IC for every bucket of channels, in order to reverse any performance drifts due to temperature. See Section 4.3 for more details.

RTOS

Due to the anticipated complexity of the firmware, especially with its multiple busses running at different speeds and interrupts from various chips, a real-time operating system (RTOS) will be used to help schedule and organize parts of the firmware with various priority levels. This way, we can move away from the regular design of a huge, very complex single main loop which checks everything. We just will need to write individual programs or "tasks" and give them a priority level. We plan to use FreeRTOS [10], which is a simple, open source RTOS that provides CPU scheduling. Being simpler than many other RTOS options, it does not provide memory management or built-in peripheral drivers, which would only complicate things, since we do not anticipate needing these. If we do need these things, we will switch to a more complex RTOS like Zephyr.

4 Technical Details

This section contains descriptions and technical explanations of each system, in ***as much detail as we currently have.*** A large amount of research and simulations, and some design and physical prototyping has already been completed, in order to verify the feasibility of these systems. Designs are subject to change.

4.1 Voltage Controlled Amplification

To implement the variable gain, voltage controlled amplifiers (VCAs) will be used, which have a gain factor related to a control voltage. The first place that gain must be remotely controlled is in the input section, which takes in either a balanced line level (preamplified) signal, or a balanced quiet microphone signal. For line level inputs, an INA2137 [11] line input receiver with fixed unity gain will impedance balance and convert the double ended signal into a single ended signal for processing. A SSI2164 [3] VCA will be used to vary the gain with remote control. For a microphone input, a PGA2500 [2] digitally controlled microphone preamplifier will be used on an SPI bus to allow control of the microphone gain. The PGA2500 does not need a control voltage, and thus does not need another control voltage DAC channel. Instead, the gain is digitally controlled through a control word sent on a serial bus.

Figure 4: Input section controls, at the top of each channel strip. Input knob controls gain applied by the VCA. The switches at the top select the source type. (MIC switches the input to pass through the PGA2500 preamplifier, and HI-Z switches in a resistor to increase the input impedance to accept direct input instrument signals from things like electric guitar coil pickups. The buttons on the bottom control the +48V phantom power for condenser microphones, phase polarity invert, and high pass filter, respectively. These functions are easy to implement with the PGA2500, but they are luxuries and will only be added if we have time.

The second place gain will need control is on the auxiliary send section, where variable amounts of audio from each channel may be summed to auxiliary signal buses for group external processing. These busses will be fed by VCAs, specifically the SSI2164 [3]. The third place for voltage controlled gain is the feed into the main summing bus. For this, two VCAs (SSI2164s) will be used per channel. One VCA will feed the left channel summing bus, and the other VCA will feed the right channel summing bus. A pan (stereo panorama balance) knob will control the difference in gain between the two VCAs for each channel, and the channel fader (volume control) will scale both VCAs equally.

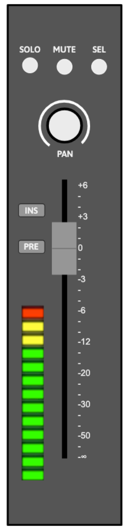

Figure 5: Output section controls, at the bottom of each channel strip. The LED strip is a simple VU meter bar display implemented with a LM3916 [12]. The channel fader is shown (See Section 4.6), and the stereo pan(orama) balance control. SOLO mutes all other channels, MUTE mutes this channel, SEL selects the channel for functions like channel grouping or group parameter changes. INS switches the channel effects insert loop, which sends the audio out, to some external processor, and recieves it back as a line-level signal. PRE sets the source point for this effects loop to be before the application of the fader and pan VCA gain. (PREfader)

Component: PGA2500

The PGA2500 digitally controlled microphone preamplifier [2] accepts its gain setting as a number, so it needs no control voltage DAC. A 16-bit control word travels over a serial port using the SPI protocol. The gain range runs from 10 dB to 65 dB in 1 dB steps, with a separate unity gain setting. Several devices may share one port in a daisy chain, which keeps the wiring count low across a large number of channels. The device also provides four programmable digital outputs. These outputs drive the pad, the phantom power supply (for condenser / capacitance microphones), the high-pass filter, and the polarity reverse relay.

The zero-crossing detector delays each gain change until the input signal crosses zero. This limits the glitch energy from the switched gain network, preventing pops and clicks on the output. A channel with no input signal therefore shows some delay before the new gain takes effect. Preamplifier recall is exact, since a switched resistor network sets the gain, and no control voltage takes part.

Component: SSI2164

The SSI2164 [3] provides four VCAs in one package. Each VCA takes a current input, produces a current output, and accepts a ground-referenced control port. The control law is exponential:

$$V_C = -\,(33\,\mathrm{mV/dB})\,G, \tag{1}$$

where G is the gain in decibels. A control span of $-660\,\mathrm{mV}$ to $+3.3\,\mathrm{V}$ therefore covers the full range of $+20\,\mathrm{dB}$ to $-100\,\mathrm{dB}$. We can just cap the lowest control voltage to 0V, so we do not need CV amplifiers. We will still need a buffer.

Noise on the control port becomes gain modulation, so the control path needs care. Noise of $1\,\mathrm{mV}$ RMS produces sidebands about $55\,\mathrm{dB}$ below the signal. Our DACs have a nominal noise voltage of 290μV peak-to-peak [6] so as long as we are careful to not add much more noise, it will not be an issue. See Section 4.9 for more information on the control voltage converters.

4.2 Voltage Controlled Resistors

Parts of the channel need a variable resistance and not a variable gain. The equalizer corner frequencies and the filter Q are the main examples. We plan to use operational transconductance amplifiers (OTAs) to implement this voltage controlled resistance, specifically the LM13700 [5], for its output Darlington buffers and linearizing diodes, which help it achieve a THD of $< 1\%$. For the LM13700 OTA [5], transconductance follows the amplifier bias current:

$$g_m = \frac{I_{\mathrm{ABC}}}{2V_T}, \qquad R_{\mathrm{eq}} = \frac{1}{g_m} = \frac{2V_T}{I_{\mathrm{ABC}}}, \tag{2}$$

where $V_T = kT/q$ is the thermal voltage. A bias current from $1\,\mu\mathrm{A}$ to $500\,\mu\mathrm{A}$ gives a resistance from about $51\,\mathrm{k\Omega}$ down to about $100\,\Omega$. Three decades of range cover a wide sweep of one corner frequency. To achieve that though, we will have to amplify the control voltages, mapping their small 0-3.3V range to something near $+/-15V$. The VCAs do not need this amplification.

A standard, real resistor will convert this buffered and amplified voltage into the biasing control current. Transconductance follows bias current in a linear way, so corner frequency also follows the control DAC code in a linear way. A lookup table in the control processor maps a target frequency to the correct code. The analog circuit therefore needs no exponential converter.

The OTA stays linear only for small differential input voltages. In most existing designs, a resistive voltage divider holds the signal at the OTA input below about $20\,\mathrm{mV}$ peak. The linearizing diodes extend this limit. We will do both.

Component: LM13700

The LM13700 [5] is a dual operational transconductance amplifier, manufactured originally by National Semiconductor, and now TI. Unlike an op-amp (which is a voltage controlled voltage source), OTAs produce an output CURRENT proportional to the differential input voltage, just like a resistor. The scaling factor (the effective resistor value) is set by an external amplifier bias current: $gm = I_ABC/2V_T$, or roughly 19 mS per mA of bias at room temperature. OTAs have been widely used in music circuits, most commonly in synthesizers, in voltage controlled filters.

Each of the two channels adds linearizing diodes and a Darlington buffer. The Darlington buffer helps drive loads at the output without loading the real output of the OTA which can affect its performance. The diodes take advantage of linearizing distortion, which predistorts the signal in a way such that the OTAs nonlinearities will undo this distortion.

The bare differential pair inside the OTA follows a hyperbolic tangent:

$$I_{\text{out}} = I_{\text{ABC}} \tanh\left(\frac{V_{\text{in}}}{2V_T}\right). \tag{3}$$

This curve stays linear only near the origin, which is why the bare part needs a small input signal. The linearizing diodes place the inverse of that curve in front of it. Two diode-connected transistors split a bias current I_D around the signal current I_s, so the voltage across the pair follows a logarithm:

$$\frac{V_D}{2V_T} = \frac{1}{2}\ln\left(\frac{I_D/2 + I_s}{I_D/2 - I_s}\right) = \text{artanh}\left(\frac{2I_s}{I_D}\right). \tag{4}$$

The differential pair then applies tanh to this result, and the two functions cancel exactly:

$$I_{\text{out}} = \frac{2I_{\text{ABC}}I_s}{I_D}. \tag{5}$$

The output is therefore linear in I_s, and the thermal voltage cancels with the curvature. Both functions come from the same junction physics on one die, so the cancellation holds across temperature. The signal current must stay below $I_D/2$, because one diode starves at that limit and the logarithm fails.

4.3 Calibration and Temperature Sensitivity

The gain constant in Equation (1) and the thermal voltage in Equation (2) both scale with absolute temperature. This is the largest threat to recall accuracy in the channel.

If we use the linearizing diodes provided for each OTA on the LM13700 chip (mentioned just above), we can greatly reduce the temperature dependency of the OTAs (Equation (2)), leaving the SSI2164 VCAs as the main problem.

The gain constant of the SSI2164 rises with absolute temperature. A rise of 30 °C moves the constant from about 33 mV/dB to about 36 mV/dB. A control voltage stored for 40 dB of attenuation then produces about 36 dB. The error grows with distance from unity gain and falls to zero at 0 dB. The thermal voltage V_T drifts in the same manner. The equivalent resistance rises by about 0.33 % per °C, and any corner frequency set by that resistance falls by the same fraction.

We correct this drift in the following way. A resistor with a +3300 ppm/°C coefficient scales the control voltage at each SSI2164 control port. If needed, we will also add a temperature sensor to the board. The control processor would read it and and trim the control DAC codes to account for the different temperature. This sensor correction could handle the residual error and the slow drift of the voltage reference.

Recall Sequence

Reconfiguring the console's state follows a fixed order. This order prevents transients at the output.

1. Ramp the channel VCA to full attenuation.
2. Write the routing, pad, polarity, and phantom power states.
3. Write the gain word to the PGA2500.
4. Write the control DAC codes for the voltage-controlled resistors.
5. Wait for the control filters to settle.
6. Ramp the channel VCA to the stored control voltage.

4.4 State-Variable Filter Equalizer

State Variable Filter Core

Each band uses a state variable filter. The topology of a state variable filter is a loop of one summing amplifier and two integrators. The output of the second integrator returns to the summing input. Three useful outputs appear at different points in this loop. The summing amplifier's output gives a high-pass response, the first integrator's output gives a band-pass response, and the second integrator's output gives a low-pass response. One filter therefore supplies every shape the band needs, and a switch (DG409) selects between them.

To build a four band equalizer, the signal is sent in parallel through four state variable filters. The output from each filter, set by the audio engineer to the desired frequency, band type, and bandwidth, is summed using a configurable gain factor and placement in the circuit to switch between cut and boost, with the raw input signal. Leaving all bands' gains at 0 will not affect the input signal, no matter their frequency or Q setting.

Traditionally, a resistor and a capacitor form each integrator. However, to implement voltage controlled center frequency, we replace the resistor with an OTA, keeping the capacitor constant and hardwired. Equation (2) of Section 4.2 gives the transconductance, so the corner frequency follows the bias current:

$$\omega_0 = \frac{g_m}{C} = \frac{I_{\mathrm{ABC}}}{2V_T C}. \tag{6}$$

Both integrators must have an identical resistance, so there will be two OTAs for frequency, but both will be controlled by one control voltage / current. A third path and OTA sets the bandwidth, or quality factor, Q. The band-pass output returns to the summing node through a damping coefficient, and Q is the reciprocal of that coefficient. The third OTA therefore makes Q variable.

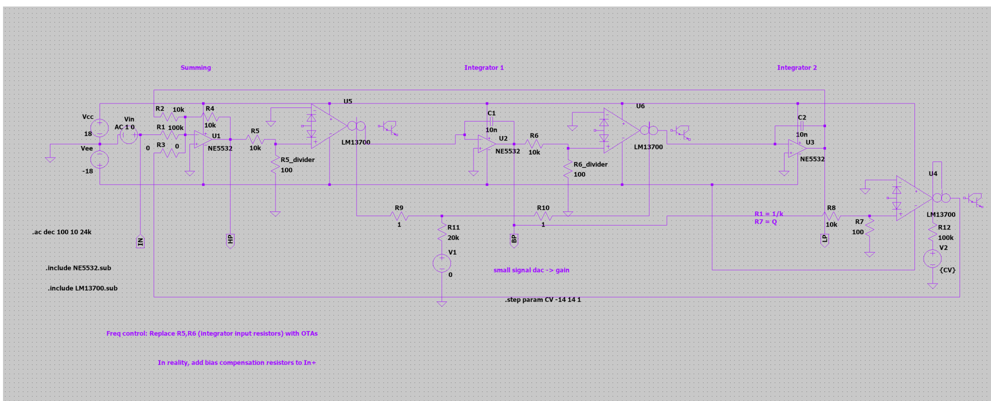

Figure 6: A single variable state filter configured with voltage-controlled center frequency and Q factor, in LTSpice.

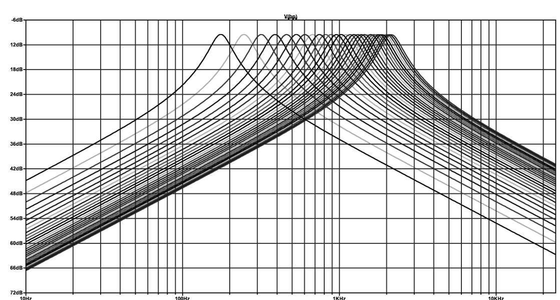

Figure 7: Output frequency response at the band pass output, during a sweep of the frequency control voltage from -14V to +14V with 1V steps. Q is kept constant. Frequency follows the same direction as voltage, so when voltage increases, so does frequency. For the other 3 in other frequency ranges, this circuit may be duplicated and the integration capacitor value changed.

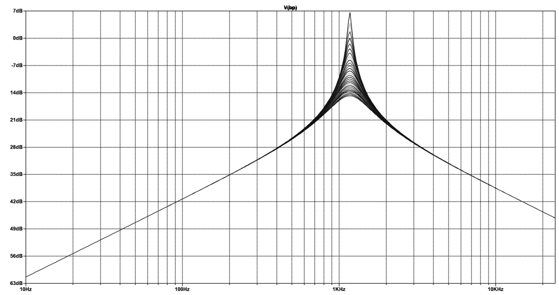

Figure 8: Output frequency response at the band pass output, during a sweep of the Q control voltage from -14V to +14V with 1V steps. Frequency is kept constant.

Band Types and Switching

The two middle bands are bells with variable frequency, Q, and gain. They take the band-pass output only, with no option to switch to shelves. The top and bottom bands are shelves that may be switched to bells, with variable frequency and gain and a constant Q. A shelf comes from the low-pass or the high-pass output of the filter circuit summed with the direct signal. A DG409 CMOS signal multiplexer selects which filter output reaches the gain stage for the top and bottom band. This part is a dual 4-to-1 multiplexer, so one device covers every mode of two bands. Its on resistance is a few hundred ohms, so each switch feeds a summing node rather than a high impedance node, which keeps the resistance out of the response. The low band selects its low-pass output for a shelf or its band-pass output for a bell. The high band does the same with its high-pass output. Further DG408 [4] devices bypass the whole equalizer and reorder it against the compressor.

Boost and Cut

Gain is the one parameter that a single OTA cannot supply well, because boost and cut of frequency bands on an equalizer need opposite signs and a reciprocal shape. The design uses one SSI2164 VCA channel per band, so one package serves all four bands. The VCA sets the magnitude of the filter contribution. In the frequency domain, using $B(s)$ for the selected filter's output and K for the VCA gain:

$$H_{\text{boost}}(s) = 1 + KB(s), \qquad H_{\text{cut}}(s) = \frac{1}{1 + KB(s)}. \tag{7}$$

The two forms are reciprocals, so a cut of 6 dB mirrors a boost of 6 dB. One analog switch chooses between them. It places the VCA output at the summing node for a boost, and inside the feedback path of the summing amplifier for a cut.

The switch never clicks, because of where it operates. The two forms meet at $K = 0$, which is the flat setting. The control processor drives the VCA to full attenuation before it moves the switch. The VCA follows an exponential law, and a boost in decibels needs a linear value of K.

A lookup table in the control processor performs this conversion. The analog circuit needs no linearizing network, which is a direct benefit of digital control.

Control Voltage Budget

Each (middle two) bell band takes three control voltages for frequency, Q, and gain. Each (outer two) shelf band take two, because their Q will be fixed. The equalizer therefore uses ten of the twenty control voltages that Section 4.9 provides for a channel. The remaining ten serve the fader, the pan control, the compressor, and the sends.

Control Surface

The panel holds four frequency encoders and four gain encoders. A potentiometer in a conventional console sits in the signal path, so each parameter needs its own potentiometer. Here, an encoder produces a number and touches no audio. The same encoder can therefore serve several parameters, and the panel carries far fewer controls than the channel has parameters.

The design uses this freedom twice. The four gain encoders carry push switches, described in Section 4.8. A held push switch turns the same encoder into the Q control for that band. The two shelf bands hold a constant Q, so their push switches select between shelf and bell instead.

Three buttons above the section form a radio group, and one lamp stays lit. They point the whole encoder group at the equalizer, at the compressor, or at the sends. Later sections describe those two processors. The encoder rings and the channel display redraw on every change of layer, so the panel always reports the parameters that the encoders now hold.

SEND	EQ	COMP
Aux 1 (press enable)	HF Gain (Hold for Q)	In Gain
Pan	Freq	Attack
Aux 2 (press enable)	HMF Gain (Hold for Q)	Thresh
Pan	Freq	Release
Aux 3 (press enable)	LMF Gain (Hold for Q)	Out Gain
Pan	Freq	Ratio
Aux 4 (press enable)	LF Gain (Hold for Q)	De-Ess Amt
Pan	Freq	

Figure 9: Since we have separated the control from the parameter being controlled, one encoder can be used for multiple functions. The audio engineer may select which part of the channel processing they are controlling with the "digital radio buttons" at the top.

A single press of a radio button selects which effect is being controlled. We also plan to add three additional control options using the existing buttons, improving flexibility and ergonomics. A double tap bypasses that processor and a second double tap restores it. A held press resets the parameters of that processor. A held press of the equalizer button and the compressor button together for three seconds exchanges the order of the two processors in the chain. The control processor mutes the channel across this last change, to prevent clicks, because it rearranges the analog path.

4.5 Dynamic Range Compressor

A compressor reduces the dynamic range of a signal: once the input rises above a set threshold, the compressor turns the gain down, so loud passages are pulled closer to quiet ones. Engineers use compressors to keep a vocal or bass sitting at a more constant volume in a mix rather than jumping forward and disappearing, and because the whole signal can be turned up to make a track sound louder and more consistent without clipping on previously loud parts, after compression is applied.

The compressor splits along the same line as the rest of the console. The detector is digital and the audio path is analog. Four high-speed analog-to-digital converters sit on either the third SPI bus (See Section 3.2), or some pins will be rearranged and the onboard ADCs used. There will be one ADC for each channel, and they report the signal to the control processor. This is the only time we will ever read the audio signal in digitally, but it is not for digital processing. The digitized signal will never be converted back; it is just to generate a control voltage envelope. From this audio signal, the processor computes the gain reduction from the threshold, the ratio, the attack, the release, and the makeup gain settings. It then drives one SSI2164 channel, and that VCA does the work on the audio.

This arrangement costs little. The converters never touch the audio path, so they need no audio-grade linearity. Only their magnitude accuracy matters. A 16-bit part gives 96 dB of detection range, which covers the working range of the compressor with room to spare. Four channels at a 48 kHz sample rate and 16 bits produce 3.1 Mbit/s, or about a third of a 10 MHz SPI bus. The detector arithmetic costs near 11 % of the processor.

The Sidechain Needs Its Own Path

One constraint governs the design. The control voltage of the compressor must not travel through the shared converters of Section 4.9. That path updates at 400 Hz and smooths its output with a 100 Hz pole. The attack time would then never fall below about 4 ms, and no setting could recover it.

The sidechain therefore carries its own converter, on the same SPI bus as the analog-to-digital converters and at the same sample rate. A reconstruction pole near 5 kHz follows it, which gives a time constant of 32 μs. The whole detection loop stays on one fast bus, and the slow control plane sets only the five parameters.

The audio path holds no delay, so the detector cannot look ahead. The gain reduction always arrives a little after the event that caused it. This suits a program-dependent leveling compressor well. It does not suit a peak limiter, and the paper does not claim one.

What Software Detection Adds

Several features arrive at no extra cost in parts. The sidechain filter is a few lines of code, so the detector can ignore low frequencies without an analog network. The detector law is a choice rather than a circuit, so peak and RMS behavior both become settings. A soft knee and a program-dependent release cost nothing further. Additionally, the console may be set up to feed other channels' signals into the sidechain ADC, allowing sidechain linking compression between sources, allowing the engineer to for example, have the bass dynamically duck during kick drum hits. Still, the actual audio signal travels through a fully analog path, as the ADCs are only to read the levels and generate the compressor CV envelope. The gain reduction is applied by analog VCAs.

Metering also comes free. The processor already holds the gain reduction as a number, so the channel display of Section 4.7 needs no measurement circuit to draw it.

De-esser Mode

A switched crossover turns the compressor into a de-esser. A DG409 routes the audio through a first-order high-pass and low-pass pair with a corner near 5 kHz. The VCA then acts on the high band alone, and the two bands rejoin at a summing amplifier. A first-order pair sums back to a flat magnitude response, so the recombined signal stays correct when the compressor is idle. The processor also moves its sidechain filter to match, so the detector responds only to the band that the VCA controls.

The detector tap follows the chain order. Section 4.4 describes the switch that exchanges the equalizer and the compressor, and the same switch moves the point at which the analog-to-digital converter reads the signal.

4.6 Motorized Faders

Each channel carries one Alps RSA0N fader [13]. This is a 100 mm motorized slide potentiometer with a 10 kΩ linear track and a touch sense track. The fader does two jobs and only two. It reports where the hand put it, and it shows where the stored value sits. The audio never reaches the resistive track.

A conventional console passes the signal through the fader track, so a worn or dirty track produces crackle in the audio. Here, the track carries a steady direct voltage and nothing else. A noisy track produces a noisy reading, and software removes that noise before the reading becomes a control voltage. The track needs to be readable, not quiet.

Figure 10: An Alps 100mm motorized fader, part number RSA0N11M9

Position Sensing

The top of the track connects to VDDA at 3.3 V, and the bottom connects to analog ground. The wiper therefore returns a fraction of VDDA. This voltage reaches an analog-to-digital converter inside the STM32G474 [8], and one converter channel serves each of the four faders. The maximum operating voltage of the track is 10 V DC [13], so 3.3 V leaves a wide margin.

VDDA also serves as the reference for the converter, as Section 4.9 describes. The reading is therefore ratiometric. A drift of VDDA moves the track voltage and the reference together, and the reported position does not change.

A $1\,\text{k}\Omega$ resistor and a $10\,\text{nF}$ capacitor sit at each converter input. The capacitor supplies the charge that the sampling network demands, and the pair also removes noise that the motor drive couples into the wiper. The converter runs with hardware oversampling, which averages the contact noise of the track. Twelve bits across 100 mm of travel give 41 counts per millimetre, or one count every $24\,\mu\text{m}$. The mechanical repeatability of the fader is far coarser than this, so the converter never limits the result.

Motor Drive

A DRV8871 [14] drives each motor. The device holds one full H-bridge and accepts two logic inputs, so each motor uses two pins on the processor. One input carries the PWM signal and the other stays static, and the pair that carries the PWM sets the direction. Both inputs high shorts the motor windings and brakes.

The motor rail is a dedicated 10 V supply, which falls comfortably within the supply range of the DRV8871 and is the recommended motor voltage of the Alps RSA0N series. A resistor on the ILIM pin sets a hard ceiling on current, and this ceiling does not depend on the duty cap. The datasheet gives $32\,\text{k}\Omega$ for a limit of 2 A [14]. The fader motor needs far less, so the design sets the limit just above the stall current of the motor. The driver also protects itself against a short circuit and against overtemperature.

The PWM runs at 25 kHz. This frequency sits above the audio band, so the motor produces no whine that a listener can hear.

Touch Sense

A TTP223-BA6 [15] capacitive sensor watches the touch track of each fader. Its output reaches one input pin on the processor, so four pins serve the four faders.

The sensor reports whether a hand is on the fader by repeatedly charging the metal fader cap and reporting capacitance via charge time. A human has additional capacitance, so when the user touches the fader, their capacitance is added to the detected amount, and the charge time increases. This way, a knocked fader or a resting cable moves the fader, but without activating the sensor. Without the TTP223, that movement would overwrite the stored value. With the sensor, the loop treats the movement as an error, since no finger was detected, drives the fader back, and leaves the stored value alone.

Position Loop

The fader position sensing loop runs at 1 kHz on the processor. It needs no traffic on the I^2C bus, because the converter and the timers are onboard the STM32 chip. The loop watches the fader's position as it drives and actively corrects the position in a feedback loop. *The fader position is updated by the value in memory.* An error in the position of the fader in this state will not effect the actual gain value stored in memory. However, when the fader is touched, this is stopped, so the audio engineer can freely move the fader, and *the value in memory is updated from the fader position.*

4.7 LCD Displays

The knob rings, the motorized fader, and the button lamps already report position and state. They cannot report a name, and they cannot report a shape. A small display adds these two things. Each channel strip carries one Adafruit 4421 module [16], which holds a 1.54 inch IPS panel of 240 by 240 pixels and an ST7789 controller [17]. The screen shows the track name, the transfer curve of the equalizer, and the gain reduction of the compressor. Adafruit supplies this part as a bare module on a flexible cable, so each strip also needs a small connector for the 22 pin "flat-flex" ribbon cable.

One controller board drives four screens on a single SPI bus configured as transmit only (no data return from screens to controller). The four modules will share the clock line and the data line (MOSI). Each module receives its own chip select line. Two pins on the controller will be used to select the module to talk to using a two bit number. A 2-to-4 decoder generates the four chip select lines and saves two pins on the processor. The backlight of all four displays will be controlled by another pin, and pulse width modulation (PWM) will be used to vary its brightness.

Bus bandwidth sets the whole update strategy. A full frame holds 57,600 pixels at 16 bits per pixel, or 115,200 bytes. A clock of 40 MHz therefore needs 23 ms for one full frame, and 92 ms for all four screens. A screen that always redraws in full is too slow for a meter.

The ST7789 holds its own frame memory. The controller sets an address window and then writes only the pixels inside that window. The design uses this feature and divides each screen into three zones.

- The track name changes only on a rename or a snapshot recall. The controller redraws this zone on that event.
- The equalizer curve changes only when the user moves an encoder. One hand moves one encoder, so at most one channel redraws this zone at a time.
- The gain reduction meter changes all the time, so this zone stays small. A bar of 16 by 200 pixels needs 6400 bytes, or 1.3 ms. Four meters at 30 updates per second use about 15 % of

Table 1: Input allocation for one channel. Each port supplies seven usable pins, because pin 7 of each port drives outputs only.

Device	**Port A**	**Port B**
Expander 1	Encoders 1 to 3, 1 button	Encoders 4 and 5, 3 buttons
Expander 2	Encoders 6 to 8, 1 button	Encoders 9 and 10, 3 buttons
Expander 3	6 push switches, 1 button	7 buttons

the bus.

A full frame buffer for four screens would need 461 kB of memory, and the controller does not hold one. The controller renders one tile at a time into a small buffer instead. A tile of 240 by 32 pixels needs 15 kB. The controller keeps two tiles and uses DMA. The processor draws into the second tile while the DMA engine sends the first tile to the bus. It is likely that during design, we will sacrifice colors to reduce the bits per pixel, in order to increase framerate. We also plan to have a single, larger display in the center / master section, for the user to configure settings and save and load console states.

4.8 GPIO Expansion and Control Hardware

Every control on a channel strip reaches the processor through the same three MCP23017 expanders [9] that share the I^{2}C bus with the converters. Each expander carries 16 pins, but two of them cannot serve as inputs. Microchip advises that GPA7 and GPB7 drive outputs only, because those pins react to noise and can corrupt the bus [9]. Each expander therefore offers 14 input pins, and three expanders offer 42.

The channel needs exactly 42. Ten EC11 encoders [18] use two pins each for their phases. Six of those encoders also carry a push switch. Sixteen buttons take the rest. The budget closes with nothing spare, so any later control needs a fourth expander. The six output-only pins remain available for indicators.

Pin Allocation

Allocation is not free choice. The expander latches a whole port at one instant, so the two phases of one encoder must sit in the same port byte. Phases split across two ports, or across two devices, would arrive from two different reads. The decoder would then see a transition that never happened. Table 1 keeps every encoder inside one port.

Encoder Signals

The EC11 is an incremental encoder with quadrature output. It holds two contact sets that open and close a quarter cycle apart. One phase leads the other, and which phase leads gives the direction of rotation. The encoder reports no absolute position and no rate. It reports only steps and direction.

The processor decodes each encoder with a small state machine. The two phases form a 2-bit value, and one detent produces one full cycle of that value. A valid step changes exactly one bit. A table of 16 entries maps the previous value and the new value to a count of $+1$, -1, or zero. A change of both bits at once cannot occur on a real rotation. The state machine treats that case as a lost sample, discards it, and resynchronizes on the next valid step. This rule matters more than raw speed. A fast turn may lose a count, but the machine never produces a count in the wrong direction.

Contact Debounce

Mechanical contacts bounce for one to five milliseconds. The expander cannot help with this on its own. Its internal glitch filter removes pulses shorter than 150 ns [9], which suppresses electrical noise but not a bouncing contact. Every input therefore passes through an RC network first.

Each input uses a $10\,\mathrm{k\Omega}$ pull-up to the rail and a $1\,\mathrm{k\Omega}$ resistor in series with the contact. A capacitor to ground completes the network. The series resistor also limits the discharge current through the contact to about 3 mA, which extends contact life. The GPIO pins of the MCP23017 are Schmitt trigger inputs. They switch at 0.2 VDD and 0.8 VDD [9], so the hysteresis band covers nearly 2 V at a 3.3 V rail. A slow edge through that band produces one clean transition and no repeat.

Two capacitor values serve two needs. Buttons use 100 nF, which gives an edge of about 1.8 ms and removes all normal bounce. Encoder phases use 22 nF, which gives an edge of about 0.4 ms. The encoder needs the faster network because its phases change far more often than a button does. A fast turn of three revolutions per second on a 20 detent encoder produces 240 transitions per second, or one transition every 4 ms.

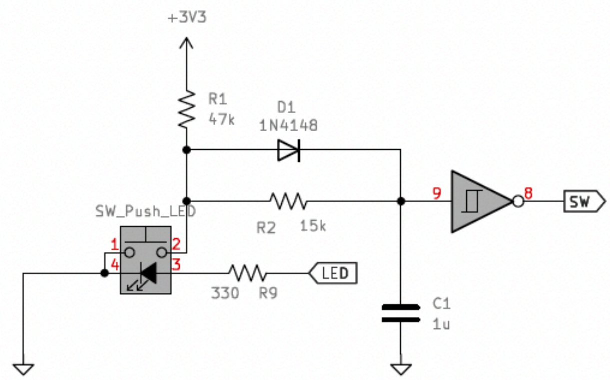

Figure 11: A debounce circuit for push buttons, with debounce time about 15ms. The Schmidt trigger is already present on the MCP23017 input pins, so an additional one does not needed to be added. We would plug node 9 straight into the MCP23017.

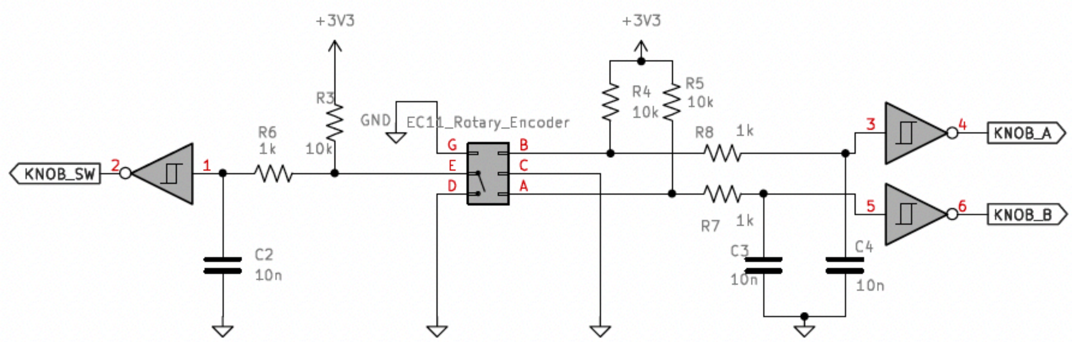

Figure 12: A debounce circuit for the EC11 rotary encoder, with both rotary step debouncing (KNOB_A and KNOB_B) and debouncing of the push switch (pressing the knob down, KNOB_SW). The Schmidt triggers are already present on the MCP23017 input pins, so additional ones do not needed to be added. We would plug nodes 1, 3 and 5 straight into the MCP23017.

Interrupt Handling

A scan of every input on every frame would waste the bus. The expanders raise an interrupt instead, and the processor reads only after a control moves. Each device enables interrupt-on-change on all input pins through GPINTEN. INTCON stays clear, so each pin compares against its own previous value.

The three devices share one interrupt line. Two configuration bits make this possible. IO-

CON.MIRROR joins INTA and INTB inside each device, so one pin reports both ports. IOCON.ODR makes that pin an open-drain output. The three pins then join at one node with a single 4.7 kΩ pull-up, and any device can pull the node low without a conflict. One line therefore serves a whole channel, and the processor uses four lines in total.

The shared line does not name its source, so the service routine scans. It reads both ports of all three devices, which takes about 370 μs at 400 kHz. It compares each byte against the stored previous value, and the changed bits identify the control. The routine then feeds the changed encoder phases to the state machine.

A read of the port registers also clears the interrupt inside each device. This step is not optional. A device that nobody reads holds the shared line low and blocks every later interrupt on that channel. The routine therefore reads all three devices on every event, and it checks the line again before it exits.

This design costs nothing while the console sits still. A fast turn of one encoder raises 240 events per second, and each event costs 370 μs. The input traffic therefore reaches about 9 % of the bus at its worst. The allowance in Table 2 of Section 4.9 treats this traffic as continuous, so that budget states an upper bound rather than a normal load.

Knob Position Indication

These encoders may control different parameters at different times or need to be set to display a different value than before during a state load. To do this, we use knobs with no indicator marking and rotary encoders with no stop point (360 degrees around, forever). A ring of 32 LEDs in small 0201 SMD packages surrounds each knob to display the position Each set set of 32 LEDs covers 270 degrees, which is the standard range of real potentiometers used in audio. One LED therefore covers about 8.7 degrees, and the ring resolves the travel to one part in 32. The ring gives a coarse reading that the eye takes in at a glance. The channel controller carries the exact value (and maybe the exact value can be shown on the channel display).

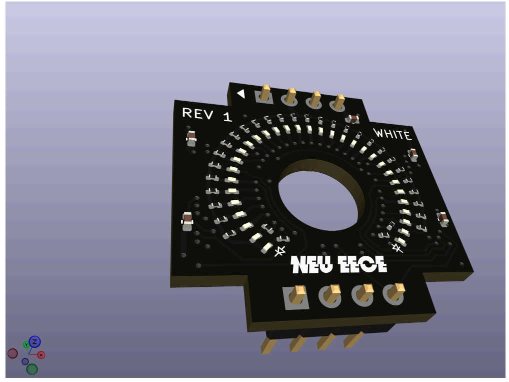

Figure 13: 3D Render of the LED ring, for the small 6mm shaft EC11 encoders, with a 7.5mm diameter hole for that shaft.

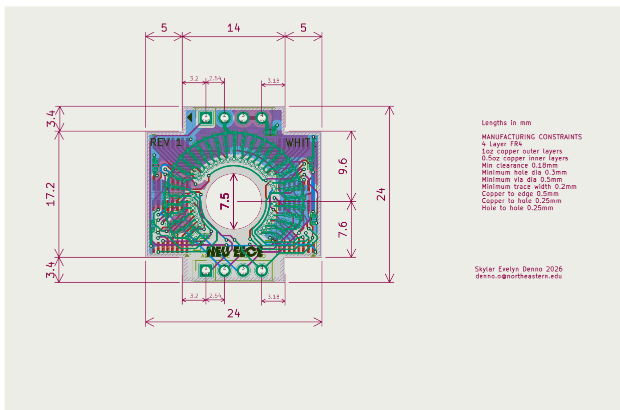

Figure 14: PCB layout. Inner layers are ground plane and VCC=5V plane. The width of these boards is only 24mm!

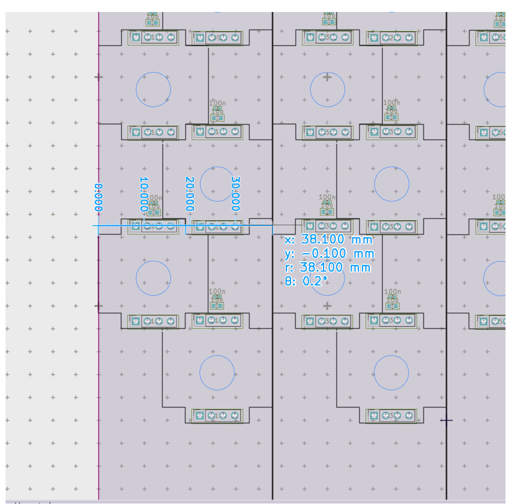

Figure 15: Aiming for a standard channel strip width of 1.5 inches (38mm), we may stagger the LED boards like this to fit knobs in two columns.

Four 74HCT595 shift registers [19] drive the 32 LEDs of one ring. They sit on a small dedicated board behind the front panel, with the encoder through the middle. Each board carries a connector for power, ground, data/clock in, latch, output enable, and data out (to cascade the bitstream to the next knob's shift registers). There are 10 knobs per channel, so forty knobs serve four channels, so 160 shift registers sit in the chain. Eight further devices on the main board drive the 64 button lamps of the four channels controlled by the bucket's controller. The chain therefore holds 168 devices and 1344 bits.

The choice of the HCT family is due to the following reason. The processor drives 3.3 V logic, and these devices run from the 5 V LED rail. An HCT input switches at 2.0 V, so it comfortably accepts a 3.3 V logic signal as high, while running at 5V [19]. The regular "HC" variant running at 5V sees logic high only beyond, 3.5 V, which would not work with 3.3V logic.

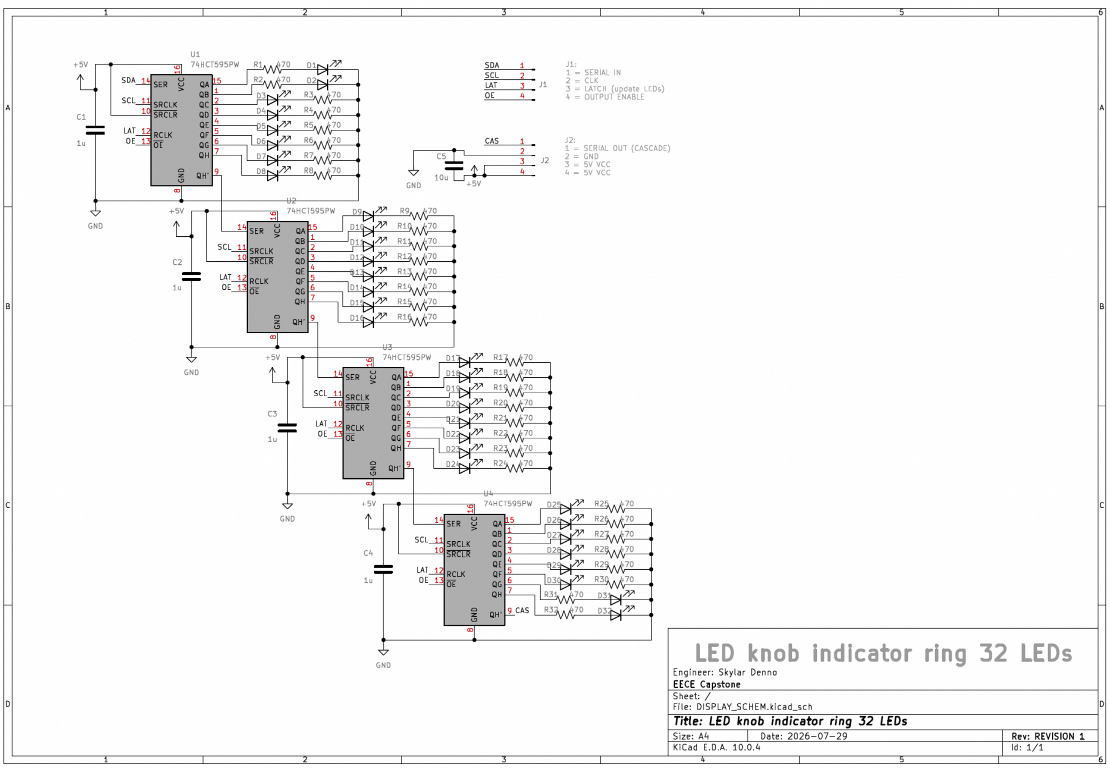

Figure 16: Schematic for the LED ring breakout boards, showing four 74HCT595 LED driver shift registers (32 total LEDs). Common cathode configuration.

The shift clock runs at 480 kHz, and a full frame of updating all LEDs therefore takes 2.8 ms. This rate is safely far below the maximum speeds the devices can handle. This is because the chain crosses the whole control surface and passes every channel strip. A slow clock with slow edges radiates far less into the analog path than a fast one. The outputs hold their state between frames, so a slow clock costs no brightness. It costs only 2.8 ms of delay before a movement appears, which is not noticeable to the human eye.

Table 2: Bus load for one channel at a 400 Hz frame rate. Each figure includes an allowance for the start, stop, and bus free periods.

Transaction	**Bytes**	**Time**	**Total**
MCP4728 fast write, 5 devices	9	210 μs	1050 μs
MCP23017 port read, 3 devices	5	123 μs	369 μs
Traffic per frame			1419 μs
Frame period at 400 Hz			2500 μs
Bus load			57 %

A dedicated 5 V rail feeds the LEDs. At 1.5 mA per LED, a fully lit surface draws about 2.0 A, so the rail carries the load and the 3.3 V logic supply does not. A PWM signal on the output enable line sets the brightness of the whole chain. That signal runs above 20 kHz. A slower PWM would place a large switching current inside the audio band, next to the analog circuits.

4.9 Control Voltage DACs

Each console channel needs many control voltages at once. These are the VCA control ports and the equalizer corner frequencies. An STM32G474 [8] supplies these levels through four independent I^2C buses. Each bus carries one console channel, so one processor serves four channels.

Each bus holds eight devices. Three MCP23017 expanders provide the switch and lamp lines described above. Five MCP4728 converters [6] provide the control voltages. The MCP4728 holds four 12-bit outputs, so each bus produces 20 control voltages. One processor therefore produces 80 control voltages in total.

Every MCP4728 leaves the factory with the same default address, so the five devices on a bus need distinct addresses. The address bits live in on-chip EEPROM. A programming step at the end of manufacture writes them once. This step needs a precise transition of the LDAC pin against the I^2C clock [6], so the test fixture performs it at a reduced clock rate. The LDAC pin serves a second purpose during normal use. The processor writes all five devices and then pulses LDAC, and all 20 outputs of a channel move together.

Bus Bandwidth

The bus runs in Fast mode at 400 kHz [7]. One bit period is therefore 2.5 μs, and one byte takes 22.5 μs because each byte carries an acknowledge bit. Two transactions repeat every frame. The Fast Write command of the MCP4728 loads all four channels in nine bytes. A two-register read of the MCP23017 returns both ports in five bytes. Table 2 gives the budget.

The bus therefore carries the full update in 1.42 ms of a 2.5 ms frame. The remaining 1.08 ms absorbs the lamp writes, the retries, and the occasional configuration write. A saturated bus would sustain about 700 frames per second, so the 400 Hz rate keeps a margin of nearly two to one. The bus stays at 400 kHz even though the MCP4728 accepts 3.4 MHz, because the MCP23017 shares the same wires and stops at 1.7 MHz. The measured rate already meets the need, and a slower clock produces slower edges and less radiated noise.

A frame rate of 400 Hz sets a Nyquist limit of 200 Hz. No control voltage can carry a component above that frequency. This limit costs nothing, because no parameter in the channel needs to move that fast.

Reference, Span, and Output Stage

Each MCP4728 takes its reference from VDDA. This is a dedicated 3.3 V linear supply that feeds the converters and nothing else. A ferrite bead and local decoupling isolate every device on the rail. The converter output is therefore ratiometric to VDDA, and this suits the design. The analog-to-digital converters share the same rail, so a measured fader position and the control voltage derived from it scale together.

The internal 2.048 V reference of the MCP4728 offers no advantage at this supply voltage. It gives a full scale of only 2.048 V at unity gain, and the gain of two would ask for 4.096 V from a 3.3 V rail. A reference of VDDA therefore gives the widest span available. A full scale of 3.3 V across 12 bits gives a step of 0.81 mV at the converter output.

A TL074 stage [20] follows each output. The stage applies a gain and an offset, and a reference voltage at the second input sets that offset. Two resistors per channel therefore fix the span. The widest span reaches about ± 13 V, because the TL074 does not swing to its rails. A design that needs a true ± 15 V output must run the stage from ± 18 V rails.

The span is not the same for every destination, and this matters. Twelve bits spread across the full 26 V output range give a step of 6.3 mV, or 0.19 dB at a VCA control port. That step is too coarse for a fader. Destinations that need fine resolution therefore receive a narrow span. A VCA control port spans -0.66 V to $+3.3$ V, which needs a stage gain of 1.2. One step at that port moves the control voltage by 0.97 mV, or 0.03 dB. Bipolar destinations such as a pan control or a filter offset use the wide span, where a coarse step does no harm.

The TL074 adds an input offset voltage of a few millivolts, and the stage gain multiplies it. On a narrow span this offset produces a fixed gain error of a few tenths of a decibel.

Reconstruction and Slew

A capacitor across the feedback resistor of each TL074 stage forms a single pole near 100 Hz. This pole does three jobs. It attenuates the images that the zero-order hold places at the frame rate. It limits the noise bandwidth of the stage, so the amplifier contributes well under 1 μV in the audio band. It also slows every parameter change.

The image rejection deserves an honest figure. A single pole at 100 Hz attenuates a 400 Hz component by about 12 dB, which is modest. The step size therefore carries the rest of the argument. The control processor limits how far a code may move in one frame, so a large parameter change arrives as many small steps. Large residual steps appear only during a fast deliberate move, and the move itself masks them.

The slew limit is a choice and not only a constraint. Slower changes in parameters usually sound more natural and musical, and they prevent artifacts that may result from sudden edges in the control voltage. The filter therefore serves the sound of the console as well as its noise budget.

4.10 Power Supply

The console holds 16 channels and a master section. Four controller cards serve those channels, and each card carries four of them. Two supplies feed the frame, and they never share a path. A linear supply feeds the analog circuits. A switching supply feeds everything digital.

Analog Supply

A toroidal transformer delivers 18 V RMS to a full bridge rectifier. Reservoir capacitors follow, and an LM317 [21] and an LM337 [22] regulate the result to ± 16.5 V. Further filtering follows each regulator. Every card carries its own regulator pair, which keeps the drop across each device small and isolates one card from another.

The analog load per channel comes from three groups. Five LM13700 packages hold the equalizer filters. Two SSI2164 packages hold the equalizer gain stages, the compressor, and the level control. About ten TL074 packages hold the control voltage output stages, the filter summing amplifiers, and the current-to-voltage converters. Together these draw near 130 mA from each rail. A card of four channels therefore needs about 520 mA per rail, and a supply of 500 mA per card meets that figure.

A separate low-dropout regulator produces VDDA at 3.3 V. This rail serves only as the reference for the converters of Section 4.9, and it draws about 50 mA. It takes its input from the linear side and not from a switching converter. A regulator of this type rejects little at the switching frequency, and VDDA sets every control voltage in the console.

Digital Supply

A 105 W switching supply feeds three buck converters. One produces 3.3 V for the processors, the expanders, the converters, and the displays. One produces 10 V for the fader motors. One produces

Table 3: Load on the 105 W switching supply. The three worst cases do not last long and do not normally coincide.

Rail	Current	Power	Condition
3.3 V digital	2.0 A	7 W	Continuous
5 V indicators	8.1 A	40 W	Every LED lit
10 V motors	5.0 A	50 W	All 16 faders start
Total		97 W	Worst case

5 V for the indicators.

The indicator load is the largest. Each controller drives 1280 ring LEDs and 64 button lamps, so the frame holds 5376. Each LED drops 2.8 V and takes 1.5 mA. A fully lit surface therefore draws 8.1 A, or 40 W from the 5 V rail. The LEDs dissipate 23 W of that figure and the series resistors dissipate the remaining 18 W. A lower rail would waste less, but 2.2 V across the resistor is what holds the current steady against the spread of forward voltage.

Table 3 shows the margin is thin. The motor figure is the reason. Sixteen motors that start together draw near 50 W, and that peak lasts only as long as the faders travel. Two measures hold the total below the limit. The ILIM resistor of each DRV8871 caps the current of one motor, as Section 4.6 describes. The control processor also staggers the starts across a recall. This second measure costs nothing that the user can hear, because the control voltage already carries the recalled value before the faders move.

Separation

Noise control rests on layout more than on components. The linear supply and the switching supply occupy separate areas of the frame, and their returns meet at one point only. The 5 V rail switches 8 A at the PWM rate of the indicators, and the 10 V rail switches several amps at the PWM rate of the motors. Both returns stay away from the analog ground.

Every device carries local decoupling. Ferrite beads isolate the supply pin of each converter and each reference, so a current transient on one device does not reach its neighbor.

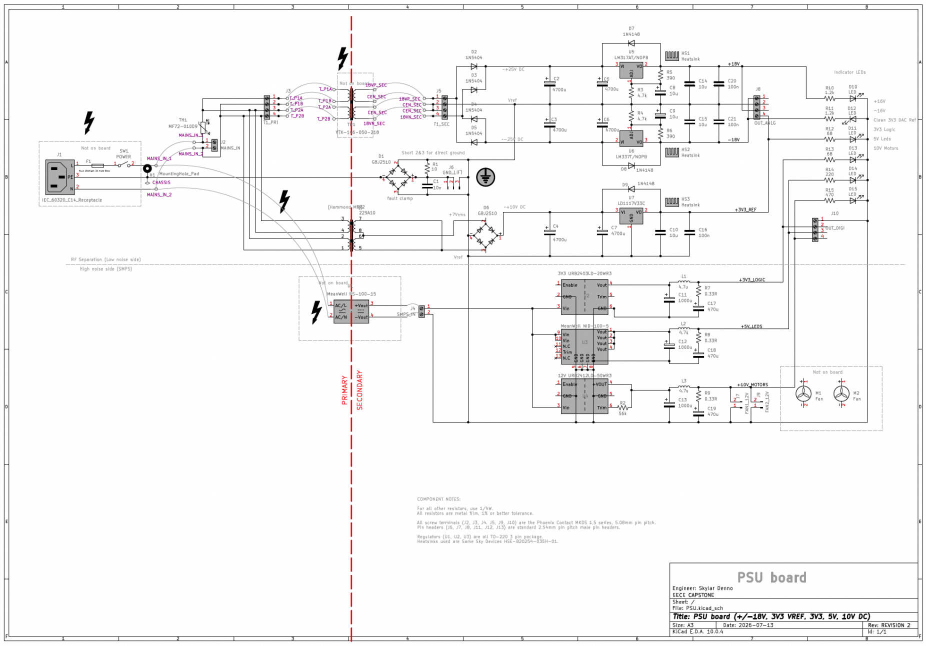

Figure 17: Schematic for the PSU. Subject to safety evaluation and change.

5 Division of Tasks

Six members divide into five subteams. Most members belong to two subteams, so no part of the console depends on one person alone. Skylar Denno directs the project and joins three subteams. This console fails at the boundaries between its parts rather than inside any single block, so one member tracks the analog design, the firmware, and the board layout at once and keeps the three in agreement.

5.1 Digital Architecture Design

Evan Forcucci, Xinhao Chen, and Daniil Petrov own this area. They chose the STM32G474 for its analog integration and its pin count, and they decide what each bus carries. The processor holds four I^2C buses and three SPI buses, and the assignment of devices to those buses sets the bandwidth budget of Section 4.9. This subteam also produces the pin assignment of Section 3.2, the control voltage converters, and the channel displays of Section 4.7. One rule guides the work. The converters read the fader position and the compressor sidechain only, and the audio itself never enters the digital domain.

5.2 Firmware

Daniil Petrov and Skylar Denno write the firmware. The task is coordination rather than computation. The processor must track the physical controls, hold the analog state, and refresh the surface, all without a delay that the engineer can feel. FreeRTOS divides this work into tasks with separate priorities, which replaces one large main loop. One processor manages a four-channel bucket, and the master section processor collects and distributes the console state for a save or a recall. The firmware also owns the interrupt service routine of Section 4.8, the recall order of Section 4.3, the fader position loop of Section 4.6, and the calibration tables of Section 4.3.

5.3 Analog Systems Design

Luke Garrity and Skylar Denno design the analog path. This subteam carries the hardest requirement in the project, because the remote control must stay transparent. A listener should not hear that a control voltage sets the gain instead of a hand. The work covers the operational transconductance amplifiers that supply every variable resistance, the voltage controlled amplifiers that supply every gain, the state variable filter equalizer of Section 4.4, and the compressor of Section 4.5. Three targets govern the design: low distortion through every gain stage, a noise floor that matches a conventional desk, and a recall that repeats without drift as the room warms.

5.4 Control and UI Design

Xinhao Chen and Evan Forcucci connect the surface to the processor. The subteam works in two stages. Software control comes first, so a working console exists before any knob reaches it. Physical controls then arrive on top of that base. The hardware covers the MCP23017 expansion of Section 4.8, the rotary encoders and their LED rings, the buttons and lamps, and the motorized faders of Section 4.6. The detailed problems live here: contact debounce, quadrature decoding, the shift register chain, and the layer scheme that lets one encoder serve the equalizer, the compressor, or the sends.

5.5 PCB and CAD Design

Daniel Salas and Skylar Denno produce the boards and the panel. Noise is the largest problem in this project, and layout decides most of it. The subteam therefore designs two kinds of board to two different standards. The analog boards need ground planes, local decoupling, and filtered supply rails. The digital boards need controlled impedance and crosstalk control at a 170 MHz clock. Section 3.1 describes the mezzanine stack that separates the two, and Section 4.10 describes the supply separation that depends on the same layout work. The subteam also designs the front panel, so that the console reads clearly to an engineer who has never used it.

6 Budget Proposal and Expense Enumeration

The budget covers one complete 16 channel console and its master section. The bucket structure of Section 3.2 makes the figure scale in a predictable way, because a console is four identical cards plus one master. A reduction to 12 or 8 channels removes whole cards and does not change the design.

6.1 Component Selection and Sourcing

Most parts come from the JLCPCB turnkey stock. This choice ties the bill of materials to the assembly service, so the same vendor supplies the part and places it on the board. Turnkey sourcing also removes the shipping cost and the customs delay of a separate order for each line. Four parts fall outside that stock. The Orient Display 5 inch master screen comes from Digikey, the THAT1646 [23] line driver comes from Mouser, the Adafruit 4421 channel screens come from Adafruit, and the LM13700 comes from a private source.

Table 4 gives the cost by subsystem for the full build and for two reduced builds. The microphone input stage carries its own row because it is the single most expensive block and the first candidate for removal. Sixteen PGA2500 preamplifiers cost $205.28, which is 19 % of the whole console. A build without microphone inputs still demonstrates every claim in Section 4, because the line receiver path exercises the same equalizer, the same compressor, and the same recall mechanism.

Table 4: Component cost by subsystem, in US dollars. The reduced builds remove the microphone input stage and then remove half the channels. The fixed cost of board and panel fabrication does not fall with the channel count.

Subsystem	**16 ch**	**16 ch, no mic**	**8 ch, no mic**
Digital control	303.52	303.52	151.50
User interface	85.90	85.90	67.76
Analog signal path	231.24	231.24	131.66
Microphone input stage	205.28	—	—
PCB, panel, passives	280.00	280.00	280.00
Total	1105.94	900.66	630.92
With 10 % contingency	1216.53	990.73	694.01

The digital control cost is dominated by converters rather than by processors. Eighty MCP4728 devices cost $165.60 and 48 MCP23017 expanders cost $82.08, while the five STM32G474 processors cost only $55.00. This ratio follows from the architecture. One processor serves four channels, but every channel needs its own 20 control voltages and its own 42 input pins.

Parts Already on Hand

A large part of the hardware already exists and therefore carries no cost in this proposal. Table 5 lists these items at the unit price the project would otherwise pay. The stock covers \$527.60 of components, which is 32 % of what the console would cost without it.

Two items deserve separate mention. The LED ring boards of Section 4.8 are already designed, so the budget carries no cost for their development. One ring serves each of the 160 encoders, and those 160 boards carry 5120 LEDs in total. The team also owns several Nucleo 64 development boards that carry the STM32G474, which covers the prototyping phase before the final boards exist. The analog power supply of Section 4.10 is also built already, so the transformer, the rectifier, the regulators, and the reservoir capacitors appear nowhere in the figures above.

Table 5: Hardware already owned, valued at the unit price in the bill of materials, against the quantity the full build needs.

Item	**Needed**	**On hand**	**Unit**	**Value**
Motorized fader, RSA0N11M9	17	16	25.00	400.00
EC11 rotary encoder	160	40	1.00	40.00
Adafruit 4421 display	16	8	10.95	87.60
LED ring boards	44	44	—	fabricated
Knob and fader caps	177	62	—	—
Buttons with lamp	256	many	0.10	—
Identified value				527.60

6.2 Known Gaps and Contingency

The stock does not cover every quantity, and the bill of materials treats three lines as fully covered when they are not. One fader, 120 encoders, and eight displays still need purchase, which adds \$232.60. The LM3916 meter driver of Figure 5 also carries a quantity of 24 at a cost of zero, which adds a further \$24.00. The corrected component total is therefore near \$1362.

Four costs remain outside the figures entirely.

1. Knob caps. One cap fits each of the 160 encoders, and the bill of materials carries 45 with no unit price. A cap from the Sifam or Rogan ranges is not a cheap part at this quantity.
2. Audio connectors. Sixteen channels need input and output connectors, and the master section needs its own. No line covers them.
3. Shipping and customs. The bill of materials states this exclusion directly.
4. Rework. A first revision of a mixed-signal board of this density rarely reaches production without one respin.

A contingency of 10 % does not absorb these items. A contingency of 30 % raises the full build to \$1437.72 and covers both the identified shortfall and a reasonable allowance for the unpriced lines. We therefore request the 30 % figure and treat the microphone input stage as the first reduction if the award falls short.

6.3 PCB and CAD Services

Three lines cover fabrication, and together they total \$280.00. Panel fabrication takes \$100.00, board fabrication takes \$150.00, and small passive components take \$30.00.

The board count is larger than a single line suggests. Section 3.1 places the analog circuits and the digital circuits on separate boards in a mezzanine stack, so one channel bucket needs two four-layer boards. Four buckets and one master section therefore need ten boards in total. The LED ring boards are a separate design at 24 mm wide, and those already exist.

Assembly is the largest risk in this section. The console holds several thousand placements across the ten boards, and hand assembly is not realistic for parts in the packages this design uses. The \$150.00 line therefore assumes JLCPCB turnkey assembly at prototype quantity. A design that needs a second revision pays this cost twice, which is the main reason for the contingency requested above.

7 Timeline

The project runs from the middle of August 2026 to the fourth of December 2026, which gives 16 weeks. Three tracks run in parallel across those weeks, and they match the subteams of Section 5. The analog track prototypes one processing block at a time. The firmware track brings up the buses and the control surface. The layout track turns each frozen design into a board.

Table 6 lists the dates that gate the other work. A design freeze matters more than a start date here, because a board cannot enter layout until its schematic stops changing.

Table 6: Gating milestones. Each freeze releases one block to the layout track.

Date	Milestone	Releases
18 Sep	Equalizer and fader design freeze	Control surface layout
25 Sep	Commitment to digital components	Controller layout
9 Oct	Input, sends, and compressor freeze	Analog board layout
16 Oct	First fabrication batch submitted	Control surface boards
6 Nov	Final designs submitted to fabrication	All remaining boards
20 Nov	Boards and panel in hand	Final assembly
4 Dec	Project due	—

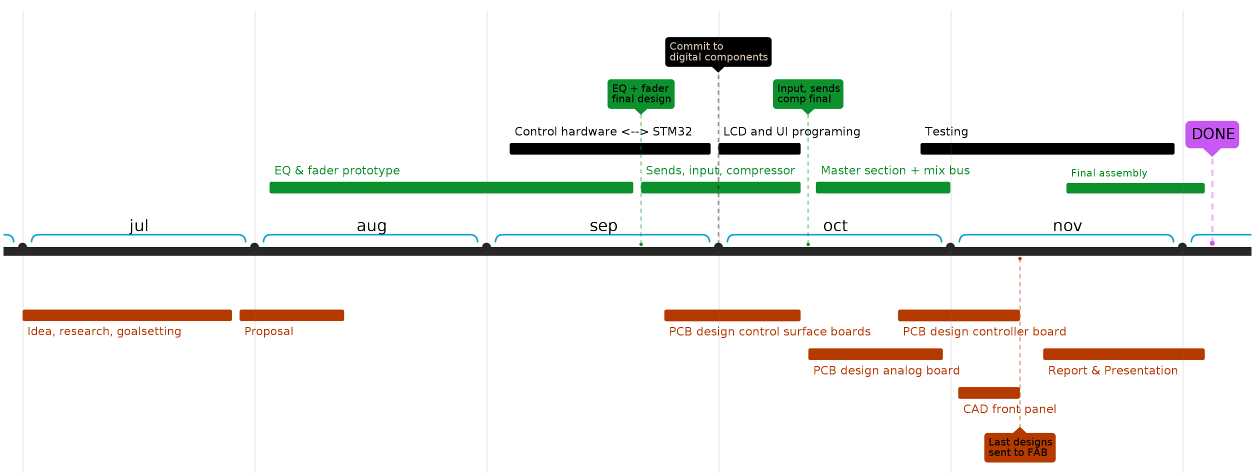

Figure 18: Projected timeline diagram - subject to changes

7.1 Design and Prototyping

This stage runs from the middle of August to the end of September. The analog subteam builds the equalizer and the fader first, because those two blocks carry the largest technical risk. The state variable filter of Section 4.4 proves that a voltage sets a corner frequency without audible cost, and the position loop of Section 4.6 proves that the motor, the touch sensor, and the control voltage agree. A design freeze on 18 September closes both.

The firmware subteam works on the link between the control hardware and the STM32 across the same weeks. The target is a working bus stack: the expanders of Section 4.8 raise interrupts, the converters accept a frame, and the encoders return counts in the right direction. A commitment to the digital component list follows on 25 September, and the controller board can then enter layout.

Two purchasing tasks belong in this stage and not later. The Alps fader is marked as not recommended for new designs, and the Adafruit display has shown long lead times. Both need an order in August, because a part that arrives in November arrives too late to matter.

7.2 Implementation and Iteration

This stage runs from late September to the sixth of November and holds most of the work. The analog subteam completes the input stage, the auxiliary sends, and the compressor of Section 4.5, and freezes them on 9 October. The master section and the mix bus follow through October. The firmware subteam moves to the displays of Section 4.7 and to the layer scheme of the control surface, then starts the test and measurement work that continues into November.

The layout track produces boards in the order that the freezes release them. The control surface boards come first, then the analog board, then the controller board. The front panel enters CAD at the start of November.

The schedule splits fabrication into two batches for one reason. A single submission leaves no room for a mistake, and a mixed-signal board of this density rarely passes on the first attempt. The control surface boards therefore go out on 16 October, three weeks before the rest. That batch is the simplest of the three, and it tests the footprints, the assembly service, and the turnkey part substitutions while a correction is still possible. The analog board and the controller board follow on 6 November.

A submission on 6 November leaves four weeks. This margin is deliberate and it should not shrink. The real constraint is not the submission date but the arrival date, because fabrication, assembly, and shipping consume two to three weeks on their own. A submission two weeks before the due date would deliver boards during the final week and leave no time to build the console around them.

7.3 Final Assembly

This stage runs from the middle of November to the fourth of December. Boards and the panel arrive near 20 November. The team populates the frame, connects the buckets to the master section, and brings up one channel at a time.

Measurement follows assembly and decides whether the project met its claim. Section 2 states the four figures that matter: total harmonic distortion, the noise floor, the crosstalk between channels, and the repeatability of a recall across a temperature range. The calibration procedure of Section 4.3 runs during this stage, because it needs the final hardware.

The report and the presentation run in parallel through November and finish with the build. Writing does not wait for the last measurement, because only the results section depends on it.

8 Conclusion

This proposal separates two jobs that a potentiometer has always performed at once. A rotary encoder now provides the physical control and the indication, and a voltage controlled element provides the resistance or the gain. Nothing in the audio path moves, and nothing in the audio path becomes a number. The console therefore recalls a session the way a digital desk does, while the signal itself never leaves the analog domain.

The architecture follows from that separation. A channel is a set of voltage controlled amplifiers and operational transconductance amplifiers, and every one of them takes its setting from a control voltage. A processor holds those settings as data, so the same encoder can serve the equalizer, the compressor, or the sends, and one command can move hundreds of parameters at once. The bucket structure repeats this arrangement four channels at a time, which lets the console scale to 8, 12, or 16 channels without a change to the design.

Three risks decide whether the approach works, and this document states each one plainly rather than assuming it away. Voltage controlled elements distort more than the fixed resistors they replace, so the design attenuates the signal at every operational transconductance amplifier input and uses the linearizing diodes to recover the loss. The control plane sits close to the audio, so the layout, the supply separation, and the reconstruction filters carry as much weight as the circuit topology. The console also runs open loop, so calibration tables and temperature correction carry the whole recall accuracy budget alone. None of these problems is solved by a better part. Each one is solved by a measurement and an iteration.

The deliverable is a demonstration and not a product. A 16 channel console with a full complement of microphone preamplifiers costs $1,105.94 in components, and existing hardware covers a further $527.60 that the budget does not carry. The bucket structure gives a clear reduction path if the award falls short, because the microphone input stage is 19 % of the cost and removing it changes nothing that the project sets out to prove. The schedule reaches fabrication in early November and leaves four weeks for assembly and measurement.

The claim under test is narrow and it is falsifiable. Digital control of an analog signal path should cost nothing that an experienced engineer can hear. If the measured distortion, noise, and crosstalk match a conventional desk, and if a recall repeats across a working temperature range, then the claim holds. If they do not, the report will say so and will say why.

9 References

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